

Nine rectangles of equal perimeter, no two alike, in a square

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Abstract

A square can be cut into nine rectangles that all have the same perimeter and no two of which are congruent, for instance an 84×84 square into pieces of perimeter 130. Nine is the least number of pieces for which this is possible, and up to rotation, reflection and scaling there are exactly five such dissections with nine pieces. The proof is an exact enumeration over all combinatorial types of dissection, carried out by two independent programs.

1 The question

Cut a square into rectangles so that every piece has the same perimeter. Cutting it into n equal strips does this, and so does any grid of equal squares. The question is whether it can be done with pieces that are all different. The question was asked on the newsgroup `alt.math.recreational` in 2006 [8], raised again by Nandakumar [5, p. 5], who guessed that no such square exists, and asked on MathOverflow in 2018 [3]. It is a cousin of the equal-area problem, where the least number of pairwise non-congruent rectangles of equal area into which a square can be cut is seven [1].

For equal perimeters the answer is yes. Figure 1 shows an 84×84 square cut into nine rectangles, each with perimeter 130, that is, each with width plus height equal to 65. Their dimensions, in order of area, are

$$59 \times 6, \quad 10 \times 55, \quad 53 \times 12, \quad 15 \times 50, \quad 48 \times 17, \quad 21 \times 44, \quad 43 \times 22, \quad 38 \times 27, \quad 31 \times 34$$

with areas 354, 550, 636, 750, 816, 924, 946, 1026, 1054, which add up to $84^2 = 7056$. Two rectangles with the same perimeter are congruent exactly when they have the same area, since width and height are then the roots of the same quadratic. So “no two congruent” can be checked by looking at the areas, and Nandakumar’s phrasing (equal perimeters, different areas) asks for the same thing.

Once one example exists, the natural questions are how few pieces suffice and how many dissections there are. Call two dissections equivalent if one is carried to the other by a rotation, a reflection or a change of scale.

Theorem 1. *For $2 \leq n \leq 8$ no square can be cut into n pairwise non-congruent rectangles of equal perimeter. For $n = 9$ there are exactly five such dissections up to equivalence. Scaled to integer coordinates in lowest terms, they live in squares of side 72, 84, 86, 161 and 165, with common perimeters 130, 130, 130, 260 and 260 (Figure 2 and Table 1).*

The proof is computer-assisted but involves no floating point and no search heuristics. Section 2 shows why the equal-perimeter condition turns each combinatorial type of dissection into a square linear system, and Section 3 describes the enumeration and its checks.

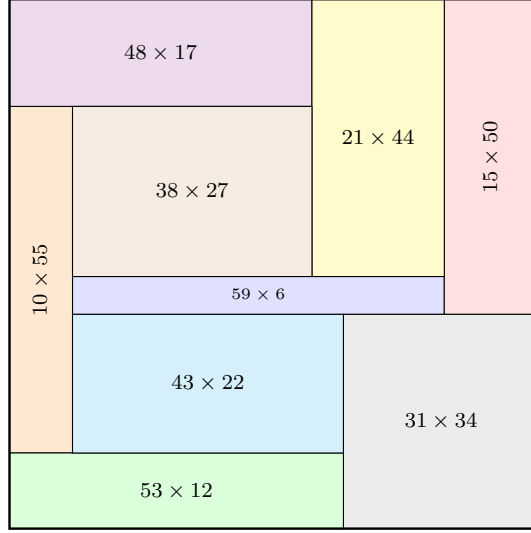


Figure 1: An 84×84 square cut into nine pairwise non-congruent rectangles, each of perimeter 130.

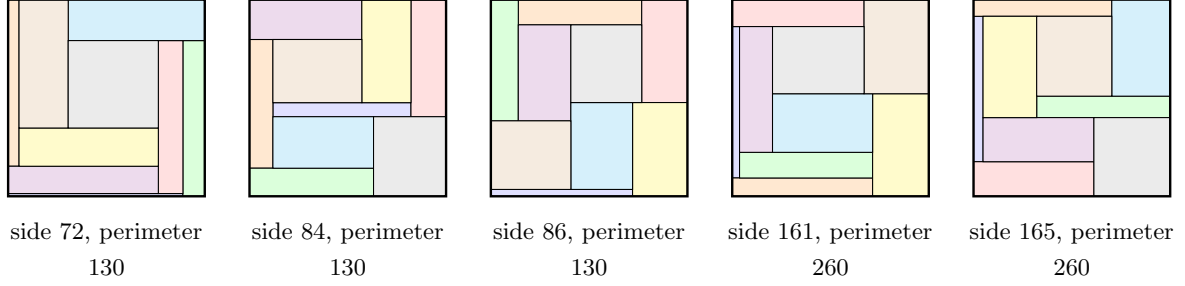


Figure 2: The five dissections of Theorem 1, each drawn at the same size.

2 One equation per piece

Start with three pieces. Cut the unit square by a vertical line at distance x from the left side, and cut the right-hand part by a horizontal line at height y , as in Figure 3. Write s for the common value of width plus height (half the perimeter). Each piece gives one equation:

$$P_1 : x + 1 = s, \quad P_2 : (1 - x) + (1 - y) = s, \quad P_3 : (1 - x) + y = s.$$

The left-hand side of each equation is the width plus the height of that piece, read off from the positions of the cuts. Three equations, three unknowns: subtracting the last two gives $y = \frac{1}{2}$, and then $x = \frac{1}{4}$, $s = \frac{5}{4}$. So this arrangement of cuts has exactly one equal-perimeter version, and in it P_2 and P_3 are both $\frac{3}{4} \times \frac{1}{2}$. The arrangement fails, and nothing else could be tried for it, because the system left no freedom.

The same thing happens in general. In a dissection of a square into rectangles, a *maximal segment* is a horizontal or vertical segment, contained in the union of the piece boundaries and not in the boundary of the square, that cannot be extended within that union. In Figure 3 there are two. The *type* of a dissection records, for each piece, which maximal segment (or which side of the square) carries each of its four sides. Two dissections of the same type differ only in the positions of their maximal segments. A point inside the square where four pieces meet is a *cross*.

Lemma 1. *A dissection of a square into n rectangles with m maximal segments and c crosses satisfies $n = m + c + 1$.*

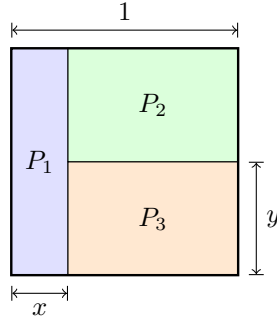


Figure 3: Three pieces. The positions x and y of the two cuts, together with the half-perimeter s , are the unknowns; each piece contributes one linear equation.

Proof. Count the corners of the pieces, $4n$ in all, by where they sit. Each corner of the square is a corner of one piece. Each end of a maximal segment lies in the interior of another segment or of a side of the square, where it forms a T and is a corner of exactly two pieces; no point is an end of two maximal segments, since two collinear ends would join into one segment and two perpendicular ends would leave a piece with a reflex angle. A cross is a corner of four pieces. Every corner is of one of these three kinds, so $4n = 4 + 2 \cdot 2m + 4c$. $\square$

Without crosses, a type has $n - 1$ maximal segments, each with one unknown position (its x -coordinate if vertical, its y -coordinate if horizontal). Fix the side of the square at 1 and add the unknown half-perimeter s . Every piece gives the equation

$$(\text{right} - \text{left}) + (\text{top} - \text{bottom}) = s,$$

where each of the four terms is the position of the segment or side that carries that side of the piece. This is n linear equations in n unknowns. When the system is nonsingular it has exactly one solution, so the type has at most one equal-perimeter realisation: the solution itself, and only if it gives every piece positive width and height. Call such a solution a *candidate*.

A dissection with crosses is covered by the same systems. At each cross, regard the horizontal maximal segment through it as two segments that meet the vertical one from either side. The result is a type without crosses in which two horizontal segments happen to share a coordinate, and the given dissection is a realisation of that type. So every equal-perimeter dissection, with or without crosses, is the solution of the system of some cross-free type.

3 The computation

To prove Theorem 1 it therefore suffices to list every cross-free type with $n \leq 9$ pieces, solve its system exactly over $\mathbb{Q}$, keep the candidates, and look among them for pieces that are pairwise non-congruent. Any candidate that survives must then be checked to be a genuine dissection. This was done twice, by programs that share no code.

Program A lists types directly. Cross-free types are known as mosaic floorplans, and their number for n pieces is the Baxter number [9]. The program builds them by repeatedly inserting a new piece at the top-left corner, in the manner of Nakano [4], and removes duplicates by a canonical form. The counts agree with the Baxter numbers 1, 2, 6, 22, 92, 422, 2074, 10754, 58202 for $n = 1, \dots, 9$.

Program B lists coordinate patterns instead. Given a cross-free dissection, nudge its maximal segments so that no two vertical ones and no two horizontal ones share a coordinate; a small enough move, with the ends of each segment sliding along the segments they meet, keeps the

type. Replacing each coordinate by its rank then gives a tiling of a $(V+1) \times (H+1)$ grid of unit cells, where V and H are the numbers of vertical and horizontal segments, by $n = V + H + 1$ rectangles, in which every interior grid line carries part of the boundary. Knuth calls these *tight pavings* [2]. The program enumerates them by repeatedly filling the lowest, then leftmost, empty cell. A type can appear several times, once for each consistent ranking of its segments; that only repeats work. The counts agree with 1, 2, 6, 24, 118, 680, 4456, 32512, 260080, the number of tight pavings with n pieces summed over grid shapes [7]. The proof of Theorem 1 rests on Program A, whose completeness is that of Nakano’s generation; Program B is an independent check, so the nudging step, stated here without a detailed proof, is not needed for the theorem.

Both programs solve every system exactly, by Gauss–Jordan elimination over $\mathbb{Q}$ in Program A and fraction-free Bareiss elimination in Program B. Both find that *every* system with $n \leq 9$ is nonsingular. Both then compare the positive solutions as point sets, under the eight symmetries of the square, with the side scaled to 1.

The number of inequivalent candidates found, with congruent pieces allowed, is

$$1, 1, 2, 6, 16, 50, 177, 664, 2532 \quad (n = 1, \dots, 9)$$

in both programs. This agrees with the sequence computed earlier and independently by Beeksmā and Rieseboš [6], which is a check on the whole pipeline: a missing type or a wrong equation would almost certainly change a count. Among these candidates, those with pairwise non-congruent pieces number 0 for $2 \leq n \leq 8$ and 5 for $n = 9$. A third script, which uses neither enumeration, checks directly from the integer coordinates in Table 1 that each of the five is a dissection of the square (pieces inside it, disjoint interiors, areas summing to the square), that all perimeters agree and that no two pieces are congruent. This completes the proof of Theorem 1.

The nonsingularity of every system is an observation, not a theorem. It is what makes each type contribute at most one dissection, and it was checked for each type rather than proved.

side	perimeter	pieces, as $[x_1, x_2] \times [y_1, y_2]$
72	130	$[0, 4] \times [11, 72], [0, 55] \times [1, 11], [0, 64] \times [0, 1], [4, 22] \times [25, 72], [4, 55] \times [11, 25],$ $[22, 55] \times [25, 57], [22, 72] \times [57, 72], [55, 64] \times [1, 57], [64, 72] \times [0, 57]$
84	130	$[0, 10] \times [12, 67], [0, 48] \times [67, 84], [0, 53] \times [0, 12], [10, 48] \times [40, 67],$ $[10, 53] \times [12, 34], [10, 69] \times [34, 40], [48, 69] \times [40, 84], [53, 84] \times [0, 34],$ $[69, 84] \times [34, 84]$
86	130	$[0, 12] \times [33, 86], [0, 35] \times [3, 33], [0, 62] \times [0, 3], [12, 35] \times [33, 75],$ $[12, 66] \times [75, 86], [35, 62] \times [3, 41], [35, 66] \times [41, 75], [62, 86] \times [0, 41],$ $[66, 86] \times [41, 86]$
161	260	$[0, 6] \times [15, 139], [0, 108] \times [139, 161], [0, 115] \times [0, 15], [6, 33] \times [36, 139],$ $[6, 115] \times [15, 36], [33, 108] \times [84, 139], [33, 115] \times [36, 84],$ $[108, 161] \times [84, 161], [115, 161] \times [0, 84]$
165	260	$[0, 8] \times [29, 151], [0, 101] \times [0, 29], [0, 116] \times [151, 165], [8, 53] \times [66, 151],$ $[8, 101] \times [29, 66], [53, 116] \times [84, 151], [53, 165] \times [66, 84], [101, 165] \times [0, 66],$ $[116, 165] \times [84, 165]$

Table 1: The five dissections of Theorem 1, in integer coordinates in lowest terms.

4 Remarks

Among the nine-piece dissections, the smallest square with integer sides has side 72; with more pieces a smaller one may exist. Nandakumar’s guess that no square works [5, p. 5] came with two others, about equal-perimeter rectangles with different areas tiling a rectangle that need

not be square: that at least seven pieces are needed, and that the number of pieces may be bounded, since he knew of no example with more than ten. The method above does not settle these directly. For a rectangle the aspect ratio is one more unknown, so each type has a one-parameter family of solutions rather than a single point, and the question becomes whether some member of that family has all dimensions positive and all pieces different.

Data and code. The programs, their output and the checking script are in the folder equal-perimeter-rectangles of <https://github.com/jvvk/mathematics>.

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